

Solving and Classifying Linear Equations

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

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|---|--|
| 1. $h = 5$ | 6. both sides simplify to (infinitely many solutions) $= 2x + 8$ |
| 2. $g = 4$ | 7. constant gap, so no solution $= 18$ |
| 3. both sides simplify to (infinitely many solutions) $= 5r + 15$ | 8. $m = 4$ |
| 4. constant gap, so no solution $= 7$ | 9. (a) $k = 20$, — |
| 5. $t = 10$ | 10. (a) constant gap, so no solution $= 24$, — |
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What is verified

8 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra

8.EE.C.7 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–4 of 5 across 12 tagged checks.

1. Bike Barn: \$9 flat fee plus \$4 per hour. Wheel World: \$14 flat fee plus \$3 per hour. Solve $9 + 4h = 14 + 3h$ and state how many solutions.

Solution: Subtract $3h$ from both sides: $9 + h = 14$. Subtract 9: $h = 5$. One variable term survives, so exactly one value of h works.

$h = 5 \Rightarrow \text{one solution}$

2. Plan 1: \$12 per month plus \$6 per gigabyte. Plan 2: \$20 per month plus \$4 per gigabyte. Solve $12 + 6g = 20 + 4g$ and state how many solutions.

Solution: Subtract $4g$: $12 + 2g = 20$. Subtract 12: $2g = 8$. Divide by 2: $g = 4$. The coefficients 6 and 4 differ, so the variable does not cancel and there is exactly one solution.

$g = 4 \Rightarrow \text{one solution}$

3. Counting by groups gives $5(r + 3)$ plants; counting row by row gives $5r + 15$. Solve $5(r + 3) = 5r + 15$ and state how many solutions.

Solution: Distribute on the left: $5r + 15 = 5r + 15$. The two sides are identical expressions, so the equation is true for every value of r .

$5r + 15 = 5r + 15 \Rightarrow \text{infinitely many}$

4. Plan A: \$25 plus \$5 each week. Plan B: \$18 plus \$5 each week. Solve $25 + 5w = 18 + 5w$ and state how many solutions.

Solution: Subtract $5w$ from both sides: $25 = 18$. That is false, and no value of w can repair it — the difference between the two balances stays at $25 - 18 = 7$ dollars forever.

$$7 \neq 0 \Rightarrow \text{no solution}$$

5. Snacks cost $8(t - 2)$ dollars, supplies $6t + 4$ dollars. Solve $8(t - 2) = 6t + 4$ and state how many solutions.

Solution: Distribute: $8t - 16 = 6t + 4$. Subtract $6t$: $2t - 16 = 4$. Add 16: $2t = 20$. Divide by 2: $t = 10$.

$$t = 10 \Rightarrow \text{one solution}$$

6. A rectangle is x cm long and 4 cm wide. Solve $2(x + 4) = 2x + 8$ and state how many solutions.

Solution: Distribute on the left: $2x + 8 = 2x + 8$. Both students wrote the same perimeter formula, so every length x satisfies the equation.

$$2x + 8 = 2x + 8 \Rightarrow \text{infinitely many}$$

7. Runner A: $30 + 6t$ metres. Runner B: $12 + 6t$ metres. Solve $30 + 6t = 12 + 6t$ and state how many solutions.

Solution: Subtract $6t$ from both sides: $30 = 12$, which is false. Both runners have the same speed, so runner A keeps her $30 - 12 = 18$ m head start at every time t .

$$18 \neq 0 \Rightarrow \text{no solution}$$

8. Draining pool: $120 - 8m$ litres. Filling pool: $40 + 12m$ litres. Solve $120 - 8m = 40 + 12m$ and state how many solutions.

Solution: Add $8m$ to both sides: $120 = 40 + 20m$. Subtract 40: $80 = 20m$. Divide by 20: $m = 4$.

$$m = 4 \Rightarrow \text{one solution}$$

9. Print shop: $4(2x + 5)$ dollars. Rival: $8x + k$ dollars. (a) Find k so that $4(2x + 5) = 8x + k$ is true for every x . (b) Explain why every other k gives an equation with no solution.

Solution (a): Distribute: $8x + 20 = 8x + k$. Subtract $8x$ from both sides: $20 = k$.

$$k = 20$$

Solution (b) — instructor-judged. A correct explanation says that subtracting $8x$ removes the variable entirely, so the equation becomes a statement about constants only: $20 = k$. If k is 20 the statement is true for every x (infinitely many solutions); for any other k it is a false numeric statement, so no value of x can make it true. Accept wording such as “the x terms disappear, so all that is left is $20 = k$.” Do not accept a bare restatement of $k = 20$ with no mention of the variable terms cancelling.

10. Tank A: $60 + 9t$ litres. Tank B: $3(3t + 12)$ litres. (a) Simplify both sides of $60 + 9t = 3(3t + 12)$ and state how many solutions. (b) Explain what the simplified equation says about the two amounts.

Solution (a): Distribute on the right: $3(3t + 12) = 9t + 36$, so the equation is $60 + 9t = 9t + 36$. Subtract $9t$: $60 = 36$, which is false; the gap between the tanks is $60 - 36 = 24$ litres at every time.

$$24 \neq 0 \Rightarrow \text{no solution}$$

Solution (b) — instructor-judged. A correct explanation says both tanks gain water at the same rate — 9 litres per minute — so tank A stays exactly 24 litres ahead and the amounts are never equal. A graphical explanation earns full credit too: the two lines have equal slopes and different intercepts, so they are parallel and never intersect. Do not accept “no solution” with no reason attached.